

For SMB builders in the ZEB era

ZEB certification × smart home



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ZEB mandate — a market made by law

Smart home is **optional**, but building energy performance is now a **legal duty**
Korea first — EU · Japan · SG follow

What is a ZEB (Zero-Energy Building)? **Green Buildings Support Act Art. 17 · Enforcement Decree Art. 12 · ZEB Certification Rules**

Minimize energy use via insulation & airtightness, generate on-site with renewables — a national certification granting **Plus-Grade 5** by energy self-sufficiency or primary energy demand.

Requirements (① Energy self-sufficiency $\geq 20\%$ or ② Primary energy demand below threshold) + ③ **BEMS or remote-metering smart meters**)

ZEB mandate roadmap

2020

Public

GFA 1,000m² +
ZEB certification mandated

2023

Public

Expanded to 500m² +
Public housing 30+ units

2025~ now

Private ('25.6 · '25.12 in force)

Multi-family 30+ units
Buildings 1,000m² +
ZEB Grade 5 required

2030

Private

Expanded to 500m² +
Virtually every new build in scope

\$131B

Korea ZEB market by 2050

CERIK · Global: EU EPBD 2030 · Japan ZEH · SG Green Mark

Cert + premium — SMB builders want both

ZEB certification brings **FAR, tax & loan** benefits;
a premium smart home **differentiates sales**, yet even one is out of reach for SMB builders

Money back on certification (statutory incentives)

Incentive	Value	Legal basis
FAR & height bonus	11~15%	Green Buildings Support Act Art. 15 · Decree Art. 11 (permanent)
Acquisition tax cut	15~20%	Special Local Tax Act Art. 47-2 ('26-end sunset · renewal cycle)
Low-rate loan	~2%/yr	Energy Rationalization Fund Guidelines
Public contribution cut	Up to 15%	Housing infra contribution rules 2-2-2
Housing & Urban Fund loan cap	+20%	2026 Housing & Urban Fund plan

☒ Fixed at pre-certification (design)

☐ Pre & main cert (realized at acquisition/completion)

✳ **FAR** — fixed at design; pre-cert is the only gate · must secure before groundbreaking
Acq. tax — **accrues to the builder (original acquisition)**, buyers by transfer excluded

Why they miss out

① Costly BEMS · HEMS

BEMS·HEMS: even small buildings need **\$36K–110K build cost** + dedicated operators
No resident-facing value beyond cert reports → not a sales edge, **pure cost**

② Costly smart home build-out

Control wall pads (lighting·heating·standby) alone: **\$330–730 each**
Plus sub-controllers · home gateway · in-unit wiring · complex servers — **a full infra set of extra cost**
Full build: **\$7,300+ per unit** — ZEB gear still separate

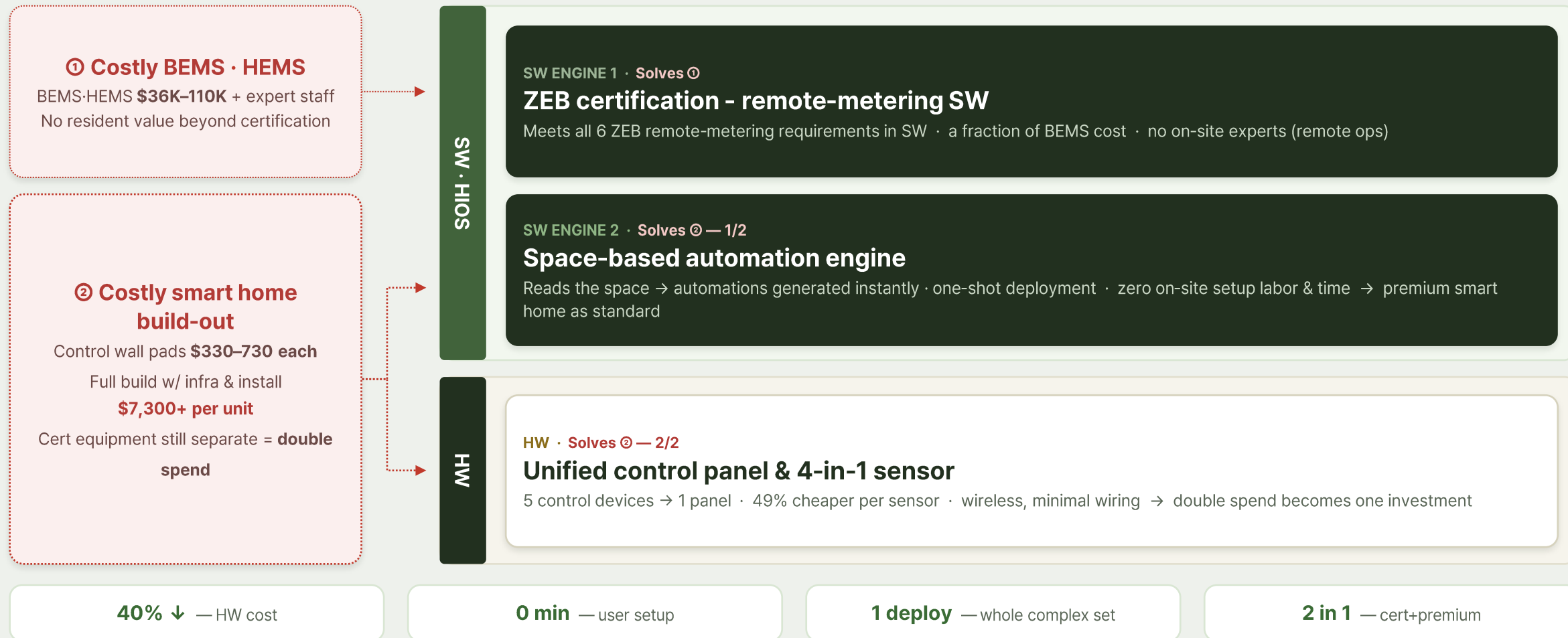
✳ **BEMS (Building Energy Management System)**

Real-time energy metering & analytics — a mandatory ZEB requirement

Cert equipment is costly with no upside; smart home gear is unaffordable — majors afford both, smaller builders neither

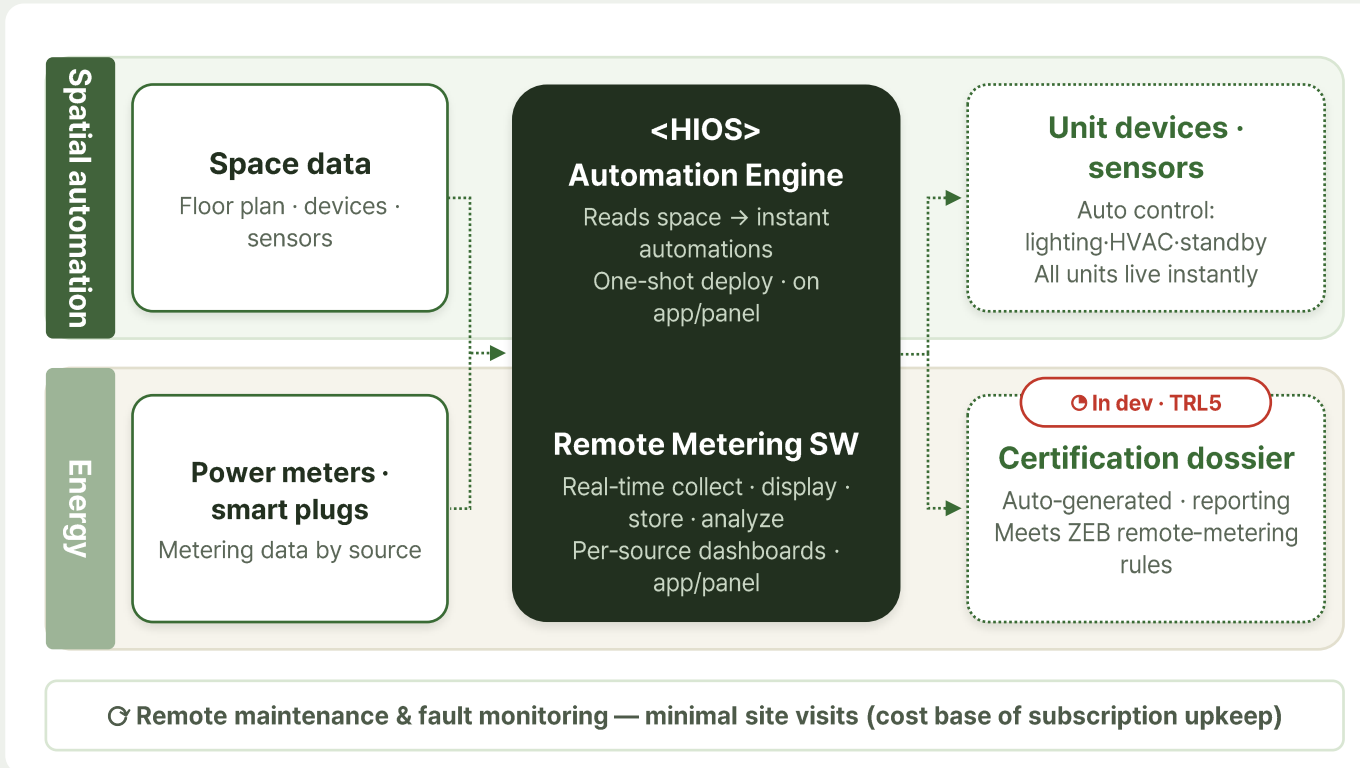
One investment, two benefits

① Certification → **remote-metering SW** · ② Smart home → **spatial automation + panel & sensor**



HIOS - automation + energy engines

Reads the space, **self-generates automations** · one deploy, every unit — setup labor & time to zero



Spatial automation

Device pre-pairing system(TRL3)

Just install — devices auto-pair with the panel · setup work zero

Automation gen. algorithm(TRL9)

Analyzes each space & devices → best scenarios generated instantly
(patent filing underway · SW copyright registered)

Energy

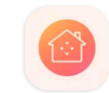
Energy data engine(TRL5)

Real-time by energy source: collect·display·store·analyze - cert compliance common base

HIO App — live on app stores

Remote control · automation setup · live status

- ✓ Released on App Store & Google Play
- ✓ Closed beta 12 users · validated at PoC sites

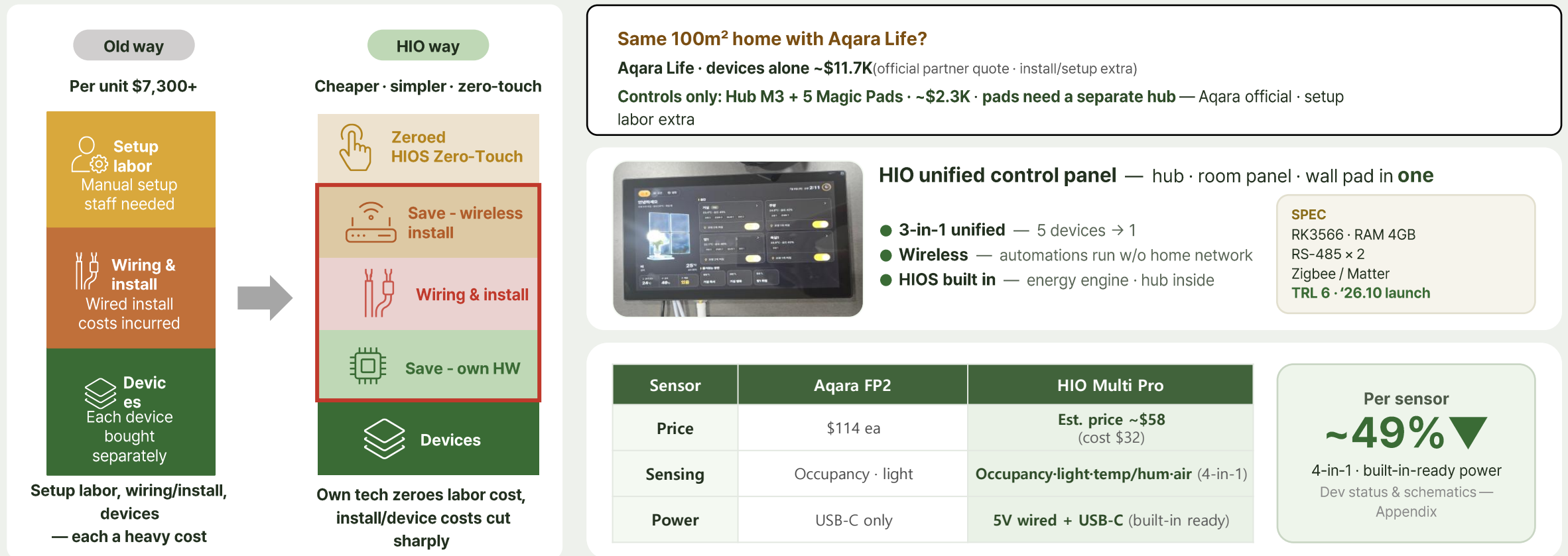


Home In One
하이오

HIOS meets all 6 required items of ZEB's SW-based energy management system (BEMS)

Smart home at ~60% of the cost

Unified control devices: 5 → 1, in-house dev cuts sensor cost ~49%



* B2C interior installs. Mandatory new-build wiring handled by licensed partner — standard install guide simplifies the process

Prices: Aqara Life official store ('26.7) — Magic Pad S1 Plus \$538 · Hub M3 \$185 · 4 pads (living + 3 rooms) · \$11.7K = official Aqara partner installer quote

Competition — two clear edges

Zero-setup automation keeps installers on their core work;
low per-unit cost captures the SMB market majors can't enter

Company	① Cert equipment compliance	② Daily use	③ Cost per unit (100m ²)	④ SMB-site batch install & deploy	⑤ Zero-Setting automation
HIOFI	Met Item3 — all 6 items	Always-on living automation	~\$2.9K devices+install incl.	Yes standard install · 1-shot deploy	Reads space → auto-generates
 Kevin LAB IoT Energy Management Service	Met core biz (large bldgs)	None admin monitoring	~\$7.3K ↑ per-unit split of \$73K+ BEMS	No price misfit for SMB	None manual control
tascon	Partial meter & lookup only	View-only usage & alert app	~\$1.5K+ (est.) wall pad & wiring — no automation	No large-complex SI projects	None monitoring only
 Aqara LIFE The IoT Solution Group	None consumer devices	Always-on rich device ecosystem	~\$11.7K+ devices only	No DIY · per-unit setup	Manual scenes built in app
 SmartThings  LG ThinQ	None appliance IoT	Always-on own appliances first	\$20K+ (est.) requires own appliances	No own builders only	Manual routines · some AI tips
 Hejhome	None	Standalone	N/A DIY items	No self-buy & install	None manual app setup

① Entry — interior firms, installers stay focused on their core work

Pre-paired package: works on install. Manual-based fitting, no training
 → **install · setup time cut by 5 days** / partner channels spread it themselves

② No head-on fight — win in a different market

Bundled 'cert+install+upkeep' deals at SMB sites sit in a **price band**
big-corp sales teams skip — a startup's niche

③ Scale — majors' weapon, standard for SMB builders

Top-tier green-smart premium delivered as a half-cost standard package

* © 100m² · devices+install, like-for-like — official prices, partner quotes & market rates (est.); feature scope differs (see cells)

Smart home — three revenue models

Enter with ①, make adoption inevitable with ②, lock in for life with ③

① Smart home build-out

Entry channel

One-off · at install

● (B2C, B2B)

Individuals · Airbnb hosts · builders · interior firms

● Sensor-driven active automation + sensor-panel-install package

● HW per unit + install

Edge Sellable to both builders & interior firms — the first market gateway

② ZEB compliance system

Inevitable adoption

Per project · around completion

● (B2B)

Developers · builders

● Metering data + cert paperwork + design partner link-up

● Fee per project (scaled to size)

Edge A mandate builders **can't opt out of** — FAR & tax perks even boost builder profit

③ Upkeep subscription

Cash flow · lock-in

Recurring (MRR) · post-completion

● (B2C)

Property mgrs · residents

● Remote upkeep + energy reporting (post-cert care)

● Monthly fee per unit (MRR)

Edge Post-cert care is a 3-year legal duty + cancel = all devices, automations & remote control stop → **structural full lock-in**

A \$197B market — opened by regulation

Not smart home gadgets — a bottom-up resize on the construction market

TAM \$197B

Korea construction market (2026 investment outlook, CERIK-RICON) · ref: ZEB \$131B by 2050 (CERIK)

SAM ~\$0.9B/yr

New residential infra — of ~430K annual housing permits ('24, MOLIT), ZEB-scope (30+ units) = ~300K units
~300K units × \$2.9K/unit (own-reference assumption) = ~\$0.9B

SOM \$1.7M

Year-3 revenue target — SMB builder segment — 20 buildings × ~\$73K each (all units) + 100 B2C units × \$2.9K
= ~\$1.75M

LAM \$73K

Path to \$73K year-1 revenue ('26) = B2C installs + early B2B contracts

① 20 B2C sites · ② Goodpeople down payment · ③ hanok & other B2B installs

TAM \$197B

SAM \$0.9B/yr

SOM \$1.7M

* A global wave — EU EPBD zero-emission new builds by 2030 · Japan ZEH · SG Green Mark; Korea is the beachhead. HIO grows on mandate penetration, not construction cycles

First revenue in · 3 full projects secured

MOUs · **paid revenue** · gov programs — demand already **validated**

01 Goodpeople + JOAN MOUs

'26.03

Key results

- **Officetel completing '27: full-project commitment** (conditional)
- **Hongdae Foundation hanok renewal** run jointly

PoC site

Samjeon-ro 7-gil 3, Songpa · 2 officetel units live

Full-project sites(3)

Samjeon-ro 9-gil 5-15, Songpa · Maegok-ri, Icheon

02 First paid revenue

'26.05~06

Key results

- **June: first B2B revenue** · tax invoice issued to JOAN
- **July: 1 more install slated**

Demand validated

2 B2C sites done as paid service, not free pilots

References built

3 real install cases for B2C traction

03 3 projects secured

'26.7~8

Key results

- **BYT HQ — breaks ground '26.8 · all 3 floors**
- **Hongdae Foundation hanok — with JOAN · documented in print**
- **Goodpeople new officetel — breaks ground '26.7 · every unit**

Why it matters

First B2B → cultural/marketing asset → large-scale — a compounding reference chain

Field records — PoC · paid installs · projects



PoC site · 2 officetel units



B2C reference · 2 units



Hongdae hanok · Before



Hongdae hanok · After (planned)

An execution-first three-person team



CEO · Business

KeonHee Lee

- Yonsei Univ. Mirae Campus, Business
- '24 Pre-Startup Pkg — top graduate ('Gapture')
- '25 G-Startup pre-track — honors
- IP Didimdol — 2 patents filed
- '26 TIPS-linked program selected
- '26 WaveLab · Modu Startup finalist
- Hackathon & competition grand prizes
- Ran a property rental business



CTO · Technology

Hogun Kim

- Kyung Hee Univ., Math & Software Convergence
- Ex-researcher, AIMS LAB
- Sole developer of HIO App & HIOS
- Designed certification data pipeline
- DL-based sleep-stage & pattern analysis
- KSC 2024 paper competition award
- SW copyrights · open-source contributor
- Certified: Big Data · Info Processing · SQLD



CPO · Product

Woojae Jang

- Yonsei Univ., Electrical & Electronic Eng.
- Ex-researcher, Yonsei FEDL lab
- Next-gen semiconductor (TFT/RRAM) R&D
- ICD internship — industry field experience
- Analog circuit design (ECG readout)
- Licensed electrical engineer
- Leads on-site install & equipment
- Semiconductor process · Python data tools

Advisors



Yunguk Kim Goodpeople GM · construction advisor

Active builder's procurement & site insight — insider view for B2B growth



Seongjin Kim Energy technology advisor

Hands-on energy equipment & metering — cert compliance know-how